📘 Chapter 37- WAN Architectures & Wireless Fundamentals

🌐 1. Introduction to WANs (Wide Area Networks)

💡 What is a WAN?

- A **WAN (Wide Area Network)** is a network that covers a **large geographical area** 🌍
- It connects **multiple LANs (Local Area Networks)** across cities, countries, or even continents

🏢 Why WAN is used?

- To connect **offices, data centers, and branches** of a company
- Helps in **data sharing and communication** between distant locations

🌎 Internet vs WAN

- The **Internet** itself is a WAN
- But in real-world usage, WAN usually means:
  - A company's **private network connections**

🔐 WAN over Internet

- Companies use **VPN (Virtual Private Network)** to create **secure connections over public internet**
- Makes communication **private and encrypted**

📜 Important Note

- Different WAN technologies exist
- Some are **old (legacy)** but still used in some countries

IN India Example

- Early internet infrastructure in India was supported by private efforts (like industrial pioneers funding connectivity)

🔌 2. WAN Over Dedicated Connection (Leased Line)

📌 What is a Leased Line?

- A **leased line** is a **dedicated connection between two sites**
- It is like renting a **private road only for your company** 🚗

⚙️ Features

- Uses **PPP or HDLC protocols**
- Provides **direct and secure communication**
- Installed via **ISP infrastructure (not direct cable end-to-end)**

💰 Why Companies Don't Buy It?

- Very **expensive** 💸
- High **maintenance cost**
- Long **installation time**

🏗️ Topology: Hub-and-Spoke

- **Hub** = main location
- **Spokes** = branch offices connected to hub

📍 Example:

- Pune = Hub
- Delhi = Spoke
- Both connected via leased line through ISP network

📉 Problem

- Slow speeds and high cost → now replaced by **Ethernet WAN & VPNs**

🌍 3. WAN Over Shared Infrastructure (Internet VPN)

🔐 What is VPN?

- VPN creates a **secure tunnel between sender and receiver**
- Data travels in **encrypted format** 🔒

🚇 Tunnel Concept (Easy Example)

- Like a **road tunnel**:
  - You can't see inside from outside 👀
- Same with VPN:
  - Data is hidden and protected

📦 Before VPN

- Data travels as **plain text**
- Anyone can read it ❌

🔒 After VPN

- Data is **encrypted**
- No third party can read it ✔️

🔄 Types of VPN

🏢 1. Site-to-Site VPN

📌 Definition

- Connects **two sites (office to office)** over internet

⚙️ How it Works

1. Data is **encrypted** 🔐
2. Encapsulated with **new headers**
3. Sent through VPN tunnel

4. Receiver **decrypts data**

🧠 **Key Points**

- Tunnel exists between **two routers**
- End devices don't create VPN themselves

📍 **Example**

- Office A router ↔ Office B router

⚠️ **Limitations**

- Doesn't support **multicast/broadcast**
- So routing protocols like **OSPF** don't work directly

💡 **Solution**

- Use **GRE over IPSec**

🔥 **Real Tool Example**

- Palo Alto firewall supports Site-to-Site VPN

🧑‍💻 2. Remote-Access VPN

📌 **Definition**

- Allows **individual users (laptop/mobile)** to connect to company network

🔐 **Technology Used**

- Uses **TLS (Transport Layer Security)**
  (same as HTTPS 🔒)

⚙️ **How it Works**

- Install VPN software (e.g., Cisco AnyConnect)
- Device connects to company router/firewall
- Creates secure tunnel

🏠 Real-Life Example

- Work From Home 💻
- Laptop connects securely to office network

🔥 Popular Tools

- Cisco AnyConnect
- Palo Alto GlobalProtect

⚖️ Site-to-Site vs Remote-Access VPN

Feature	Site-to-Site 🌐	Remote Access 💻
Use Case	Office to Office	User to Office
Technology	IPSec	TLS
Users	Multiple devices	Single device
Connection	Permanent	On-demand

✂️ Split Tunnel VPN

📌 What is it?

- Only **important traffic goes through VPN**
- Other traffic goes through normal internet

💡 Example

- Office server access → VPN
- YouTube, Games → Normal Internet

🎯 Benefits

- Saves **bandwidth** 📉
- Reduces **cost** 💰

- Improves performance ⚡

🌐 4. MPLS VPN (Multi Protocol Label Switching)

📌 **What is MPLS?**

- A WAN technology using **labels instead of IP addresses**

🏗️ **Devices**

- **CE Router** → Customer Edge
- **PE Router** → Provider Edge
- **P Router** → Provider Core

⚙️ **How it Works**

- PE router adds **labels** to data
- Data is forwarded using labels (not IP)

🔄 **Types**

🔹 **Layer 3 MPLS**

- CE ↔ PE use routing protocols (e.g., OSPF)
- Routes are shared

🔹 **Layer 2 MPLS**

- CE routers act like directly connected
- No routing with PE
- Same subnet

🎯 **Benefit**

- Faster and efficient routing

🌐 5. Internet Connectivity

📡 Types

- Leased Line
- MPLS
- DSL
- Cable (CATV)
- Fiber Optic (most popular now ⚡)

📶 6. Wireless Fundamentals

📡 Standards

- Defined by **IEEE 802.11**
- Certified by Wi-Fi Alliance

📌 Important

- Wi-Fi is a **brand name**, not a standard

📡 7. Wireless Network Challenges

⚠️ Issues

1. All devices receive all signals 📡
2. Privacy concerns 🔐
3. Collision problems

⚙️ Solution

- Wired: CSMA/CD
- Wireless: **CSMA/CA** (avoid collisions)

📡 8. Radio Frequency (RF)

⚡ How Wireless Works

- Electric current → antenna → electromagnetic waves

📊 Key Terms

🔹 Frequency

- Cycles per second (Hz)

Unit	Meaning
Hz	1 cycle/sec
kHz	1,000 cycles
MHz	1 million
GHz	1 billion

🔹 Period

- Time for one cycle
- Example: 4 Hz = 0.25 sec

📡 9. Wi-Fi Frequency Bands

🔹 2.4 GHz

- Longer range 📶
- Lower speed 🐢
- More interference ❌

🔹 5 GHz

- Short range
- High speed ⚡
- Less interference ✔️

🚴 **Example**

- Slow cycling → long distance (2.4 GHz)
- Fast cycling → short distance (5 GHz)

📡 10. Channels

📌 **Concept**

- Bands are divided into channels

🔹 **2.4 GHz**

- Use **1, 6, 11** (non-overlapping)

🔹 **5 GHz**

- Many non-overlapping channels

🐝 **Design**

- Use **honeycomb pattern** for AP placement

📡 11. Wi-Fi Standards

Standard	Frequency	Speed	Name
802.11	2.4 GHz	2 Mbps	-
802.11b	2.4 GHz	11 Mbps	-
802.11a	5 GHz	54 Mbps	-
802.11g	2.4 GHz	54 Mbps	-
802.11n	2.4/5 GHz	600 Mbps	Wi-Fi 4
802.11ac	5 GHz	6.9 Gbps	Wi-Fi 5
802.11ax	2.4/5/6 GHz	Very Fast	Wi-Fi 6

📡 12. Signal Behavior (Very Important)

🔥 1. Absorption

- Signal converts into heat
- Weakens signal

📍 Example: Wall absorbs Wi-Fi

🪞 2. Reflection

- Signal bounces off metal

📍 Example: Elevator → weak signal

🌊 3. Refraction

- Signal bends in different medium

📍 Example: Glass, water

🌀 4. Diffraction

- Signal bends around obstacles

📍 Example: Signal reaches behind wall

🌫️ 5. Scattering

- Signal spreads in all directions

📍 Example:

- Rain 🌧️
- Dust
- Trees 🌳

🎯 Final Summary

- WAN connects **long-distance networks**
- Leased lines = **expensive but dedicated**
- VPN = **secure communication over internet**
- MPLS = **label-based fast routing**
- Wireless uses **radio waves**
- 2.4 GHz = range, 5 GHz = speed
- Signal behavior affects **Wi-Fi performance**

🧾 🔥 SUMMARY

🌐 WAN Basics

- WAN connects **multiple LANs over large distance** 🌍
- Used by companies to connect branches, data centers, offices

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🔌 WAN Types

1️⃣ Leased Line

- Dedicated private connection 🔒
- Uses PPP / HDLC
- Expensive 💸

2️⃣ VPN (Internet-based WAN)

- Secure communication over internet 🔐
- Uses encryption
- Types:
  - Site-to-Site VPN 🌐
  - Remote Access VPN 💻

🌍 MPLS WAN

- Uses **labels instead of IP** 🚀
- Faster forwarding
- Used by ISPs

📡 Wireless Fundamentals

- Based on **IEEE 802.11**
- Uses radio waves 📡

📶 Frequencies

- **2.4 GHz** → Long range, low speed
- **5 GHz** → Short range, high speed

📡 Signal Issues

- Absorption 🔥
- Reflection 🪞
- Refraction 🌊
- Diffraction 🌀
- Scattering 🌫️

🧠 Key Concept

👉 WAN = Connectivity
👉 Wireless = Mobility

🎯 ✅ CONCLUSION

💡 Modern networks use:

- VPN instead of costly leased lines
- MPLS for enterprise-level performance
- Wireless for flexibility and mobility

🔥 Final understanding:

WAN connects the world 🌍
Wireless connects users 📱

🎤 Q & A

❓ Q1: What is WAN?

✅ **Answer:**
A WAN (Wide Area Network) is a network that connects multiple LANs across large geographical areas like cities or countries.

👉 Example:
Company branches in Mumbai, Delhi connected via WAN.

❓ Q2: Difference between LAN and WAN?

Feature	LAN	WAN
Area	Small	Large
Speed	High	Lower
Cost	Low	High

❓ Q3: What is a leased line?

✅ **Answer:**
A leased line is a dedicated private connection between two locations.

👉 Key points:

- Always ON ⏱️
- Secure 🔐
- Expensive 💸

❓ **Q4: Why companies don't prefer leased lines today?**

✅ **Answer:**

- High cost 💰
- Slow deployment
- Maintenance heavy

👉 Replaced by VPN & MPLS

❓ **Q5: What is VPN?**

✅ **Answer:**
VPN (Virtual Private Network) creates a secure encrypted tunnel over the internet.

👉 Data becomes unreadable to 3rd party 🔐

❓ **Q6: What is tunneling?**

✅ **Answer:**
Tunneling means encapsulating data inside another packet for secure transmission.

👉 Like a tunnel 🚇 → no one can see inside

❓ **Q7: Types of VPN?**

✅ **Answer:**

1. Site-to-Site VPN 🌐
2. Remote Access VPN 💻

❓ **Q8: Difference between Site-to-Site and Remote Access VPN?**

Feature	Site-to-Site	Remote Access
Use	Office ↔ Office	User ↔ Office
Device	Router	Laptop
Protocol	IPSec	TLS

❓ **Q9: Why GRE over IPSec is used?**

✅ **Answer:**
Because IPSec does not support multicast/broadcast.

👉 GRE adds support for routing protocols like OSPF.

❓ **Q10: What is Split Tunnel VPN?**

✅ **Answer:**
Only important traffic goes via VPN, rest uses internet.

👉 Saves bandwidth + improves performance 🚀

❓ **Q11: What is MPLS?**

✅ **Answer:**
MPLS is a WAN technology that forwards packets using labels instead of IP.

👉 Faster than traditional routing 🚀

❓ **Q12: MPLS components?**

✅ **Answer:**

- CE (Customer Edge)
- PE (Provider Edge)
- P (Provider Core)

❓ Q13: Difference between MPLS L2 and L3?

Type	Description
L2 MPLS	Same subnet
L3 MPLS	Routing used

❓ Q14: What is Wi-Fi?

✅ **Answer:**
Wi-Fi is a wireless networking technology based on IEEE 802.11 standards.

❓ Q15: Difference between 2.4 GHz and 5 GHz?

Feature	2.4 GHz	5 GHz
Range	Long 📶	Short
Speed	Low 🐢	High ⚡
Interference	High ❌	Low ✔️

❓ Q16: What are Wi-Fi channels?

✅ **Answer:**
Channels divide frequency bands to avoid interference.

👉 2.4 GHz → 1, 6, 11

❓ Q17: What is CSMA/CA?

✅ **Answer:**
Collision Avoidance method used in wireless networks.

👉 Devices check before sending data

❓ Q18: Difference between CSMA/CD and CSMA/CA?

Feature	CSMA/CD	CSMA/CA
Use	Wired	Wireless
Method	Detect collision	Avoid collision

❓ Q19: What is absorption?

✅ Signal absorbed → becomes weak

❓ Q20: What is reflection?

✅ Signal bounces off metal

❓ Q21: What is diffraction?

✅ Signal bends around objects

❓ Q22: What is scattering?

✅ Signal spreads in multiple directions

🧠 🔹 SCENARIO-BASED QUESTIONS

❓ Q23: Company wants secure communication over internet. What will you use?

✅ **Answer:**
VPN (IPSec or SSL VPN)

❓ Q24: User working from home needs access to office network?

✅ **Answer:**
Remote Access VPN

❓ Q25: Multiple branches need constant connection?

✅ **Answer:**
Site-to-Site VPN or MPLS

❓ Q26: Wi-Fi signal weak in office?

✅ **Answer:**
Check:

- Interference
- Walls (absorption)
- Channel overlap

❓ Q27: How to improve Wi-Fi performance?

✅ Answer:

- Use 5 GHz
- Proper channel planning
- Add more APs
- Avoid interference

❓ Q28: Why only channels 1, 6, 11 used?

✅ Answer:
Because they do not overlap → reduces interference

❓ Q29: What happens if too many users on same AP?

✅ Answer:

- Congestion
- Packet loss
- Low speed

🎯 🔥 One Liner

"Explain WAN & Wireless in simple way"

✅ Best Answer:

WAN connects networks over long distance using technologies like VPN and MPLS, while wireless allows devices to connect without cables using radio frequencies like 2.4 GHz and 5 GHz. Both are essential for modern networking—WAN for connectivity and wireless for mobility.

🏁 ✅ FINAL TAKEAWAY

✔ WAN = Long distance communication 🌍
✔ VPN = Secure communication 🔐
✔ MPLS = Fast routing 🚀
✔ Wireless = Flexible connectivity 📡
✔ QoS + Wireless + WAN = Modern Network Backbone 💯
